

Weak amenability with one infinite-center rank-one factor

Automated mathematical discovery run `fa_banach_001`

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Status. Candidate substantial partial result, likely valid, subject to human review. The theorem proves the conjectured sufficiency direction for connected Lie groups having at most one rank-one simple Levi factor with infinite center. It does not settle the remaining multi-factor central mixing or the full necessity direction.

1 Source problem

The source is S. Knudby, *Weak amenability of Lie groups made discrete*, arXiv:1612.04280 [1]. On source p. 2 it recalls that the weak-amenability classification for connected Lie groups was known when the Levi factor has finite center and when the group is simply connected, and then states: “Some intermediate cases remain open.” The passage is reproduced below.

$$\Lambda(G) = \prod_{i=1}^n \Lambda(S_i).$$

Combining Theorem 3.1 and Theorem 1.1 one can then compute the value $\Lambda(G)$ for any connected semisimple Lie group G . Our proof of Theorem 3.1 relies on an inequality proved by Cowling for discrete groups (see (2.6)). In order to apply Cowling’s inequality, we pass to lattices by using Haagerup’s result (2.5) that this does not change the Cowling-Haagerup constant. The same trick is also used in our proof of Theorem 5.4.

Theorem 3.1 was previously known under the additional assumption that the semi-simple Lie group had finite center or finite fundamental group. Indeed, under this additional assumption Theorem 3.1 then follows from Theorem 1.1 and an application of the well-known permanence properties (2.1) and (2.3) below. The assumption of finite center or finite fundamental group can be considered as extreme cases, and Theorem 3.1 then settles the intermediate cases in which the center and the fundamental group are both infinite.

In a similar spirit, the characterization of weak amenability for connected Lie groups in general has been done in the cases where the Levi factor has finite center [10] or the Lie group has trivial fundamental group [25] (the case of finite fundamental group then follows from (2.1)). Some intermediate cases remain open.

More explicitly, write a Levi decomposition of the Lie algebra as

$$\mathfrak{g} = \mathfrak{r} \rtimes \mathfrak{s}, \quad \mathfrak{s} = \bigoplus_{i=1}^k \mathfrak{s}_i,$$

where $\mathfrak{r}$ is solvable and the $\mathfrak{s}_i$ are simple. The simply connected classification [2, Theorem 1.10] asserts that weak amenability is equivalent to

$$\text{for every } i, \text{ either } \text{rank}_{\mathbb{R}}(\mathfrak{s}_i) = 0, \quad \text{or} \quad \text{rank}_{\mathbb{R}}(\mathfrak{s}_i) = 1 \quad \text{and} \quad [\mathfrak{s}_i, \mathfrak{r}] = 0. \quad (1)$$

The open classification predicts the same criterion without a restriction on global topology.

We write $\Lambda_{\text{WA}}(H)$ for the Cowling–Haagerup constant. We use the standard permanence facts collected in [1, Section 2]: compact normal quotients preserve this constant; direct products multiply it; and if $N \triangleleft H$ is closed and H/N is amenable, then $\Lambda_{\text{WA}}(N) = \Lambda_{\text{WA}}(H)$. We also use the source’s semisimple theorem [1, Theorem 3.1], which in particular shows that the constant of a connected simple Lie group is unchanged on passing among its connected local forms.

2 Proof intuition

Under condition (1), every noncompact Levi factor splits away from the amenable part of the universal cover. Factors with finite center can be made adjoint by one finite compact quotient. What remains is an amenable group A and, at worst, one simple group U whose center is infinite cyclic, modulo an arbitrary discrete central subgroup D .

The key is to close the image of U . Its quotient in the ambient group is amenable, so it alone determines the Cowling–Haagerup constant. Modulo the part of D already lying in A , the kernel is cyclic. A cyclic subgroup of an abelian Lie group either is closed (unless its closure is compact) or has compact closure. In the closed case the closure of U is just U ; in the compact case it differs from the adjoint form $U/Z(U)$ only by a compact central group. Both alternatives have exactly the expected constant.

Two infinite cyclic simple centers are qualitatively different: a rank-two kernel can have dense image in a noncompact real direction. This explains both the reach and the stopping point of the argument.

3 The cyclic-closure lemma

Lemma 1. *Let A be an amenable Lie group and let U be a connected simple Lie group whose center is infinite cyclic. Let*

$$D \leq Z(A) \times Z(U)$$

be discrete, and put $Q = (A \times U)/D$. Then

$$\Lambda_{\text{WA}}(Q) = \Lambda_{\text{WA}}(U).$$

Proof. Let $q : A \times U \rightarrow Q$ be the quotient map and set $N = \overline{q(U)}$. The group N is closed and normal, and Q/N is a quotient of A , hence amenable. Therefore

$$\Lambda_{\text{WA}}(Q) = \Lambda_{\text{WA}}(N). \tag{2}$$

Fix a generator z of $Z(U)$, let

$$A_0 = \overline{\text{pr}_A(D)}, \quad D_A = D \cap (A \times \{e\}).$$

Because multiplying an element of D by U absorbs its U -coordinate,

$$q^{-1}(N) = \overline{UD} = A_0 \times U, \quad N = (A_0 \times U)/D.$$

The group D_A is discrete central in A_0 . If the projection of D to $Z(U)$ is trivial, then $D = D_A$, $A_0 = D_A$, and $N \cong U$, proving the claim.

Otherwise this projection is $\langle z^\ell \rangle$ for some $\ell \geq 1$. Choose $(a, z^\ell) \in D$. Then D/D_A is generated by the class of this element. In the abelian Lie group

$$B = A_0/D_A,$$

the cyclic subgroup generated by $b = aD_A$ is dense by the definition of A_0 . The elementary structure theorem for abelian Lie groups now gives the following dichotomy: either B is compact,

or $B = \langle b \rangle$ is an infinite discrete cyclic group. Indeed, a dense cyclic subgroup cannot be dense in a nonzero Euclidean factor; if its projection to a discrete factor is nonzero, convergence forces the exponent to be eventually constant, making the cyclic subgroup closed; with neither type of factor, the closure is compact.

If $B = \langle b \rangle$ is infinite discrete, every element of A_0 can be absorbed using a power of (a, z^ℓ) and an element of D_A . Thus the map $U \rightarrow N$ is surjective. Its kernel is trivial, since b has infinite order, so $N \cong U$.

Suppose instead that B is compact. After dividing by D_A we may write

$$N \cong (B \times U) / \langle (b, z^\ell) \rangle.$$

The image

$$C = (B \times Z(U)) / \langle (b, z^\ell) \rangle$$

is central in N and is compact: every one of its elements has a representative in

$$B \times \{e, z, \dots, z^{\ell-1}\},$$

a compact set. Moreover $N/C \cong U/Z(U)$. Compact-quotient invariance and the semisimple theorem therefore yield

$$\Lambda_{\text{WA}}(N) = \Lambda_{\text{WA}}(U/Z(U)) = \Lambda_{\text{WA}}(U).$$

Together with (2), this proves the lemma. $\square$

4 Partial classification theorem

Among the real-rank-one simple real Lie algebras, precisely those of type $\mathfrak{su}(n, 1)$, $n \geq 1$, have a simply connected Lie group with infinite center; here $\mathfrak{su}(1, 1) \cong \mathfrak{sl}(2, \mathbb{R})$.

Theorem 2. *Let G be a connected Lie group whose Lie algebra satisfies condition (1). Suppose that at most one rank-one simple ideal $\mathfrak{s}_i$ is of type $\mathfrak{su}(n, 1)$ for some $n \geq 1$. Then G is weakly amenable. More precisely, if $\tilde{S}_i$ is the simply connected simple Lie group with Lie algebra $\mathfrak{s}_i$, then*

$$\Lambda_{\text{WA}}(G) = \prod_{i=1}^k \Lambda_{\text{WA}}(\tilde{S}_i).$$

Thus the conjectured sufficiency direction and its predicted constant hold for this class.

Proof. Let $\tilde{G} \rightarrow G$ be the universal covering homomorphism and let D be its discrete central kernel. In the simply connected Levi decomposition, separate the compact simple factors from the noncompact ones. By condition (1), every noncompact factor commutes with the solvable radical and therefore is a direct factor. Consequently

$$\tilde{G} \cong A \times P \times U.$$

Here A is the semidirect product of the simply connected solvable radical by the compact Levi factors, and is amenable; P is the product of the rank-one factors having finite center; and U is absent or is the unique rank-one factor having infinite cyclic center.

Let $F = Z(P)$, a finite group, and let $K = (DF)/D \triangleleft G$. The group K is finite, hence compact. Passing to G/K does not change the Cowling–Haagerup constant. In $\tilde{G}/F$, the factor P/F is center-free. Therefore the image D' of DF/F has trivial P/F coordinate and

$$G/K \cong (P/F) \times Q, \quad Q = (A \times U)/D', \quad D' \leq Z(A) \times Z(U).$$

The group D' is still discrete, since only a finite group was removed.

If U is absent, then Q is amenable. If U is present, Lemma 1 gives $\Lambda_{\text{WA}}(Q) = \Lambda_{\text{WA}}(U)$. Hence product and compact-quotient invariance give

$$\Lambda_{\text{WA}}(G) = \Lambda_{\text{WA}}(P/F)\Lambda_{\text{WA}}(Q) = \prod_{i=1}^k \Lambda_{\text{WA}}(\tilde{S}_i).$$

The last equality uses the semisimple local-form theorem and the fact that compact factors have constant one. In particular the displayed value is finite, so G is weakly amenable. $\square$

Example 3 (A genuinely intermediate topology). *Let U be the universal cover of $\text{SU}(n, 1)$, let z generate $Z(U)$, and fix $\alpha \notin \mathbb{Q}$. In $\mathbb{R} \times U$ set*

$$D = \langle (1, e), (\alpha, z) \rangle, \quad G = (\mathbb{R} \times U)/D.$$

The subgroup D is discrete although its projection $\mathbb{Z} + \alpha\mathbb{Z}$ to $\mathbb{R}$ is dense. The group G is neither simply connected nor covered by the finite-center-Levi hypothesis; indeed the analytic image of U is injective and dense. Nevertheless Theorem 2 gives $\Lambda_{\text{WA}}(G) = \Lambda_{\text{WA}}(U)$. This shows that the theorem is not merely a reformulation of either previously known extreme case.

5 Upgrade attempts and remaining obstruction

Eight focused routes were audited. First, lifting an arbitrary lattice to the universal cover does *not* descend weak amenability: Cowling’s discrete central-quotient inequality has the opposite direction from the one needed. Second, a lattice that splits in the universal cover would work, as in the source’s free-lattice argument for $\text{SL}(2, \mathbb{R})$, but splitting is a real cohomological obstruction for complex hyperbolic lattices. Third, periodizing compactly supported Herz–Schur multipliers over D has no known uniform completely bounded norm estimate.

Fourth, the cyclic-closure route gives Lemma 1 and Theorem 2. Fifth, it cannot simply be iterated: with two infinite centers the homomorphism

$$\mathbb{Z}^2 \longrightarrow \mathbb{R}, \quad (m, n) \longmapsto m + \alpha n$$

can have dense noncompact image, neither compact nor discrete. Sixth, an attempt at necessity via locally bad analytic subgroups stops because such subgroups need not be closed in intermediate topology. Seventh, no non-weakly-amenable closed discrete subgroup was obtained from central quotients of rank-one lattices. Eighth, the later-literature search described below found no theorem removing the remaining obstruction.

Thus the packet makes no claim when two or more $\mathfrak{su}(n, 1)$ factors can mix simultaneously with a noncompact central direction. It also does not claim the necessity half in arbitrary intermediate topology. The source proves additional special cases by different lattice methods, notably multiple $\mathfrak{sl}_2$ factors at constant one; the present result is complementary, not a maximal catalogue.

6 Novelty audit and verification recommendation

A bounded search on 13 August 2026 covered arXiv:1612.04280 and arXiv:1505.00984, the exact phrase “some intermediate cases remain open”, “weak amenability connected Lie groups arbitrary fundamental group”, “reductive Lie groups lattices”, and “lattice lifts to the universal cover” for $\text{SU}(n, 1)$. It found the known finite-center, simply-connected, semisimple, and special $\text{SL}(2, \mathbb{R})$ results, plus later work on residual finiteness of complex-hyperbolic central extensions, but no full later classification or this cyclic-closure subcase. Novelty confidence is provisional.

Human review should focus on two points: the decomposition after killing $Z(P)$ in Theorem 2, and the claim that the dense cyclic group in Lemma 1 has either compact closure or is itself closed.

Both have been expanded in the proof so that no multiplier construction or lattice-splitting assumption is hidden.

References

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- [4] M. Cowling and U. Haagerup, *Completely bounded multipliers of the Fourier algebra of a simple Lie group of real rank one*, Invent. Math. 96 (1989), 507–549.